

3.8.4.**a.**

One dimension being the numbers Moonbeam crosses out, another being numbers Sunshine would cross out.

Or, you could define it as $\binom{20}{2} \cdot 2!$ since there's the numbers that Moonbeam/Sunshine choose, and then $2!$ ways to arrange between them.

b.

y can be anything, starting at 20, and x can be anything more than 2 less than y , starting at 18.

As y decreases, x also decreases the same amount since it's a linear relationship.

That means the possible values of x for each decreasing y is 18, 17, 16...1, so to find all the possible x, y combos, we add that.

The sum is $\frac{18 \cdot 19}{2} = 171$. For probability, divide by total distribution space, which is 380.

$\frac{171}{380} = 45\%.$

3.8.5.**a.**

20 has 4 pairings that work

10 has 2 pairings that work

5 has 1, and 1 has 1.

All over $8 \cdot 7$ since 8 for first choice, 7 for second.

b.

Drawing 20 means instant win, and $\frac{1}{4}$ are 20.

c.

$\binom{8}{2}$ in the total probability space.

2 pairings that are the same num: $10 + 10$ or $20 + 20$.

6 ways to form ordered pairs, 2 ways to get each pair since each bill has 2

d.

It works since it includes all the possible pairings, and if order mattered it would double both the count and the probability space.

3.9.5.**a.**

For $k = 10$, what's the chance you get exactly three rolls that add up to 8?

Chance for one dice roll to add up to 8: $2 + 6, 3 + 5, 4 + 4$, and reverse the first two so total of 5 ways over $6^2 = 36$

For exactly three, it's $\left(\frac{5}{36}\right)^3 \cdot \left(\frac{31}{36}\right)^7 \cdot \frac{10!}{3!7!}$, where we multiply by the arrangement of dice that could've been "fortunate"

b.Min value of k such fortunate roll is $> 50\%$ Just “not” getting an un-fortunate roll, which is $1 - \left(\frac{31}{36}\right)^n$, complimentary counting!

$$1 - \left(\frac{31}{36}\right)^n = 0.5$$

$$\left(\frac{31}{36}\right)^n = 0.5$$

$$n = \frac{\log(0.5)}{\log\left(\frac{31}{36}\right)}$$

$$\boxed{n = 5}$$